

Appendix B — Glossary

Key terms used throughout the course.

ADC — Analog-to-Digital Converter. Converts a voltage on a GPIO pin to a digital value. Used in Day 20 to read battery voltage via the nRF52840's built-in SAADC peripheral.

BLE — Bluetooth Low Energy. A short-range wireless protocol optimised for battery-powered devices. Used in Days 17, 18, and 26.

CBOR — Concise Binary Object Representation (RFC 7049). A binary encoding format used in Days 12 and 16 to serialise sensor data compactly over USB CDC-ACM.

CDC-ACM — USB Communications Device Class – Abstract Control Model. The USB class that makes the nRF52840 appear as a virtual serial port (`/dev/ttyACM0`) on the host PC. Used in Days 11–13, 15–16.

DTS / Device Tree Source — A hardware description file (`.dts` / `.overlay`) that tells Zephyr which peripherals exist, where they are mapped, and how they are connected. Central to every Zephyr project.

GATT — Generic Attribute Profile. The BLE protocol layer that defines services and characteristics. Used in Days 17 and 26.

GPIO — General Purpose Input/Output. Digital pins driven high/low as output or sampled as input. Used in Days 5, 9, and throughout.

I2C — Inter-Integrated Circuit. A two-wire serial bus (SDA + SCL) for communicating with sensors and peripherals at short range. Used in Days 14–17 and 25–27.

ISR — Interrupt Service Routine. A function called directly by hardware when an interrupt fires. In Zephyr, ISRs must not sleep, block, or allocate memory — use a work item or semaphore to defer to a thread.

Kconfig — The Linux-inherited build configuration system. Boolean and integer options that enable/disable subsystems at compile time. Configured via `prj.conf`.

NVS — Non-Volatile Storage. A simple key-value store on a dedicated flash partition that persists data across resets and power cycles. Used in Days 22 and 25–27.

PPK2 — Nordic Power Profiler Kit 2. A USB measurement device for sampling current consumption down to ~100 nA, used to verify sleep-state current figures.

PWM — Pulse Width Modulation. Controls average power to an output (e.g., LED brightness) by varying the duty cycle of a digital signal. Used in Day 21.

SDK — Software Development Kit. Here: the Zephyr SDK (version 1.0.1), which bundles the ARM GCC toolchain, OpenOCD, and J-Link tools required to compile and flash nRF52840 firmware.

SPI — Serial Peripheral Interface. A four-wire full-duplex bus (MOSI, MISO, SCK, CS) for high-speed peripherals such as flash memory. Used in Day 19 with the W25Q SPI flash.

System OFF — The deepest power state on the nRF52840. The CPU is fully stopped and RAM is not retained. Quiescent current is typically 0.4–0.7 μ A. Wake is via GPIO edge or reset. Used in Day 24 and the capstone (Day 27).

West — Zephyr’s meta-build tool. Manages the workspace, fetches modules with `west update`, and wraps CMake/Ninja with `west build` and `west flash`.

Work Queue — A Zephyr kernel object (`struct k_work`, `struct k_work_q`) that executes work items in a dedicated thread. Used in Day 23 to defer processing from ISR context safely to thread context.